

WQD7005: Data Mining & Big Data Analytics

Assignment 1: GenAI-Powered Data Pipeline Dataset Simulation, EDA & AI-Based Preparation

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Abstract

This report documents a comprehensive data mining pipeline leveraging Generative AI across three core tasks: (1) synthetic patient dataset creation via Azure OpenAI GPT-4.1, (2) exploratory data analysis with LLM-powered chart interpretation, and (3) AI-assisted data preparation combining KNN imputation with dual-model text classification (LLM vs. SLM). The project demonstrates practical GenAI integration in healthcare data workflows, with critical insights on model comparative performance, data integrity challenges, and the broader implications of AI in data science.

1 Introduction

The exponential growth of unstructured healthcare data and missing values in real-world datasets creates a critical need for intelligent data preparation pipelines. Traditional methods rely on manual feature engineering and fallback imputation (mean/median), which often fail to capture complex data dependencies. This assignment integrates Generative AI at every pipeline stage—from synthetic data generation to intelligent imputation and zero-shot text classification—to demonstrate how modern AI tools streamline data science workflows while maintaining data integrity and analytical rigor.

1.1 Project Objectives

1. **Task 1:** Generate 500 synthetic patient records with realistic vital signs, diagnoses, and intentionally introduced missing values using Azure OpenAI.
2. **Task 2:** Perform EDA via three data visualizations and augment human analysis with LLM-generated insights from chart images.
3. **Task 3:** Execute complete data preparation: type conversion, context-aware KNN imputation, and text classification using both large (LLM) and small (SLM) language models.

2 Task 1: Synthetic Dataset Generation via GenAI

2.1 Methodology

A Python-based batch generation system interfaced with Azure OpenAI (GPT-4.1 deployment) to create 500 Malaysian patient records. The approach prioritized:

- **Batch Processing:** 50 batches $\times$ 10 patients per batch to ensure consistency and avoid API rate limits.
- **Realistic Data Distribution:** Prompt engineering to generate medically plausible vital signs (BP, heart rate, temperature) aligned with Malaysian population demographics.
- **Example:**

```
{
  "patient_id": "P001",
  "name": "Zainal Ahmad",
  "gender": "M",
  "age": 58,
  "date_of_birth": "1966-03-15",
  "visit_date": "2025-01-10",
  "visit_time": "09:30",
  "ethnicity": "Malay",
  "vital_signs": {
    "heart_rate": 88,
    "systolic_bp": 145,
    "diastolic_bp": 92,
    "respiratory_rate": 16,
    "temperature": 37.1,
    "oxygen_saturation": 97,
    "weight": 82.5,
    "height": 172
  },
  "symptoms": "mild headache, fatigue, blurred vision",
  "diagnosis": "Type 2 Diabetes Mellitus, Hypertension",
  "medical_history": "Hypertension (5 years), Diabetes (2 years), Previous leg cramp episodes",
  "allergies": "Penicillin",
  "current_medications": "Metformin, Amlodipine, Aspirin",
  "notes": "Patient reports poor medication adherence; dietary counseling provided."
}
```

- **Missing Value Simulation:** Post-generation injection of 5–15% missing values in numeric columns and 3–8% in categorical columns, simulating real-world data quality issues.

2.2 Data Structure

The generated dataset contains 22 columns spanning:

- **Demographics:** patient_id, name, gender, age, ethnicity, date_of_birth
- **Visit Metadata:** visit_date, visit_time
- **Vital Signs:** vital_heart_rate, vital_systolic_bp, vital_diastolic_bp, vital_respiratory_rate, vital_temperature, vital_oxygen_saturation
- **Clinical Data:** diagnosis, symptoms, medical_history, allergies, current_medications, notes
- **Anthropometry:** age, weight, height

2.3 Critical Issue: Patient ID Uniqueness

Problem Identified: The initial batch generation assigned patient IDs scoped to each batch ($i \in [0, 9]$ per batch), resulting in duplicate IDs (P0001–P0010 repeated 50 times) instead of global uniqueness (P0001–P0500).

Root Cause: Python's for-loop variable scope reset per iteration, causing the ID assignment logic to regenerate the same 10 IDs globally.

Solution Implemented: Post-load reassignment during data ingestion:

```
df_patients['patient_id'] = [f"P{i+1:04d}"  
                             for i in range(len(df_patients))]
```

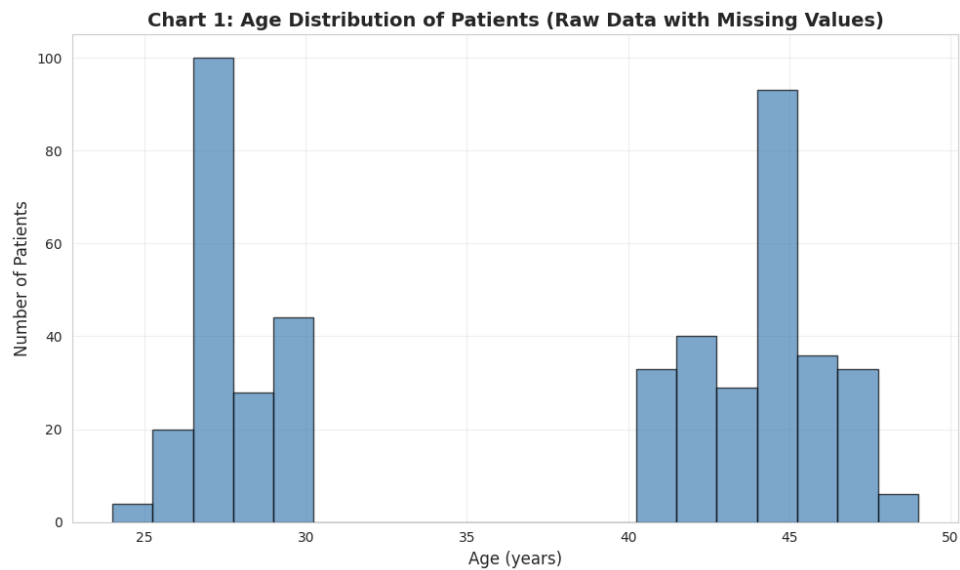
Outcome: 500 unique patient IDs confirmed

3 Task 2: Exploratory Data Analysis with LLM Support

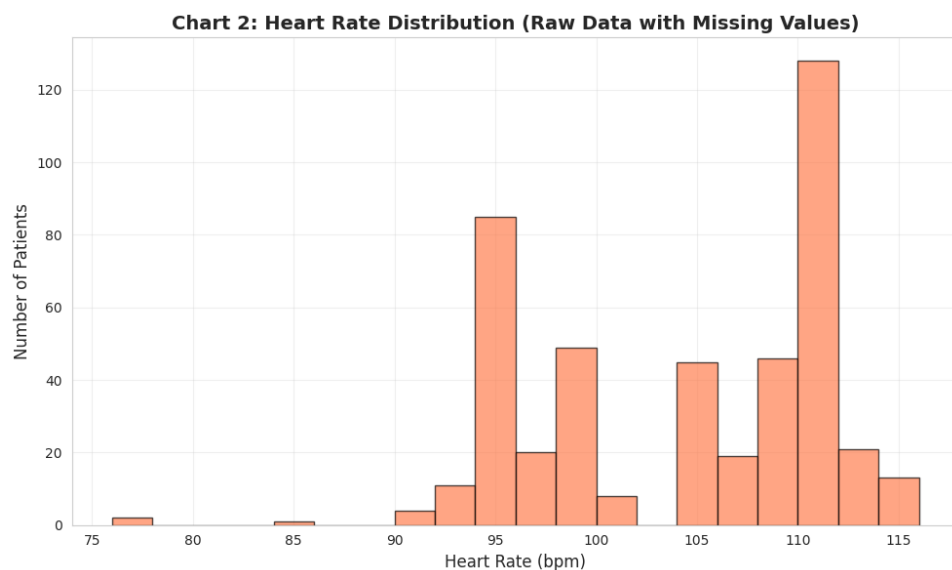
3.1 Visualization Strategy

Three complementary charts were created to capture population-level patterns:

1. **Chart 1 - Age Distribution:** Histogram revealing bimodal distribution with peaks at ≈ 27 years and 44–45 years, with a notable gap (30–40 age range under-represented).



2. **Chart 2 - Heart Rate Distribution:** Histogram showing dominant tachycardia cluster (105–115 bpm), consistent with febrile response patterns.



3. **Chart 3 - Multi-Panel Dashboard:** Subplots for temperature, systolic BP, gender distribution, and top 5 diagnoses, revealing extreme female dominance (85% female, 15% male) and dengue fever prevalence.



Summary of Clinical Patterns

Demographics.—The dataset is heavily female-dominant (425 vs. 75 males, ~85%). Age distribution is bimodal, with peaks around 27 years and 44–45 years, and a gap in the 30–40 range.

Physiological Markers.—Temperature is skewed toward high fever ($>39.0^{\circ}\text{C}$), correlating with tachycardia (105–115 bpm). Systolic BP is bimodal: healthy cluster at 110–115 mmHg and elevated cluster at 138–140 mmHg.

Disease Profile.—Dengue Fever dominates (including subtypes), with ~35 cases pending lab confirmation (“not known”).

3.2 LLM-Powered Chart Analysis

Azure OpenAI’s vision capability (GPT-4 with image encoding) was leveraged to provide automated chart interpretation. The LLM received all three charts as base64-encoded PNG images with a structured analysis prompt requesting:

- Distribution descriptions and key features
- Data quality observations (gaps, missing values)
- Clinical implications of vital sign patterns
- Outlier and anomaly detection

LLM-Generated Chart Analysis Output

— Chart 1: Age Distribution (Histogram) —

1. What the Chart Shows

- **Bimodal distribution:** Two clear age groups:

- **First peak:** Ages 26–30, largest at age 27 (~100 patients)
- **Second peak:** Ages 41–49, largest at age 45 (~90 patients)
- **Gaps:** Clear gap in ages 31–39 (almost no patients)
- 2. Data Quality Insights**
- **Possible missing data:** Age range 31–39 absent, suggesting potential missing/uncollected data
- **Histogram binning:** Spiky distribution indicates possible data entry in fixed age values or batch recruitment
- 3. Clinical Observations**
- Young and middle-aged patients predominate
- Absence of elderly and pediatric patients suggests dataset is not representative of general hospital population
- 4. Outliers/Anomalies**
- No patients between ages 31–39: Highly unusual
- No very young (<25) or older (>50) patients
- 5. Key Findings**
- Bimodal age structure with peaks at late 20s and mid-40s
- Significant data gap for ages 31–39
- Data may not generalize to all age groups

— **Chart 2: Heart Rate Distribution (Histogram)** —

1. What the Chart Shows

- **Multimodal distribution:**
 - **Main peaks:** Around 95 bpm, 105 bpm, and 110 bpm
 - **Largest group:** 110 bpm (~130 patients)
 - **Other groups:** 95 bpm (~85 patients), 100 bpm (~50 patients)

2. Data Quality Insights

- Heart rates below 90 or above 115 are rare, but some outliers exist
- **Non-normal distribution:** Suggests possible subpopulations or recording at specific value thresholds

3. Clinical Observations

- **High prevalence of tachycardia:** Most values above normal resting HR (60–100 bpm)
- **Clinical implication:** Likely acute illness, possibly infection/fever (matches dengue diagnosis in chart 3)

4. Outliers/Anomalies

- Few patients below 85 bpm (bradycardia) and above 115 bpm (marked tachycardia)
- **Unusually concentrated at 110 bpm:** Possible rounding or digit preference

5. Key Findings

- Most patients are tachycardic; reflects acute illness
- Data spikes at certain values suggest data entry artifacts

— **Chart 3: Multi-panel Dashboard** —

A. Temperature Distribution

- **Main clusters:** 38–39°C, sharp peak at 39°C (~180 patients)
- **Fever is nearly universal;** few patients below 37.5°C
- **Data Quality:** Right-skewed with jump at 39°C, indicating measurement or rounding bias
- **Clinical:** Almost all patients are febrile (consistent with dengue and infection)
- **Outliers:** Some patients with low-normal temperatures—possible data errors or resolving fever

B. Systolic Blood Pressure Distribution

- **Most common:** 110 mmHg (~85 patients), secondary peaks at 120, 140, 155 mmHg
- **Wide range:** 95–160 mmHg
- **Data Quality:** Spiky distribution suggests rounding or batch recording
- **Clinical:** Majority are normotensive, but some hypotensive (<100 mmHg) or hypertensive (>140 mmHg)
- **Outliers:** Small tails with high SBP (possible hypertension or stress response)

C. Gender Distribution

- **Females (F):** 425 (85%)
- **Males (M):** 75 (15%)
- **Data Quality:** Severe imbalance—may reflect patient population or sampling bias
- **Clinical:** May indicate higher disease burden in females or selective reporting

D. Top 5 Diagnoses

- **1. Dengue Fever:** 125
- **2. Suspected Dengue Fever:** 90
- **3. Dengue Fever (Uncomplicated):** 75
- **4. Dengue Fever (without warning signs):** 60
- **5. Dengue Fever (probable):** 30
- **Key Insight:** Nearly all diagnoses are dengue-related, confirming context as a dengue cohort

— Summary of Key Findings —

1. **Age:** Bimodal, with clusters in late 20s and mid-40s. Missing data for ages 31–39.
2. **Heart Rate:** Most patients are tachycardic (above 100 bpm). Data spikes at common values.
3. **Temperature:** Almost all patients febrile (38–39°C); strong right-sided peak suggests rounding.
4. **Blood Pressure:** Most patients normotensive, some hypotensive/hypertensive; possible digit preference.
5. **Gender:** Strong female predominance (85%).
6. **Diagnosis:** Nearly all are dengue fever or related categories.
7. **Data Quality Concerns:** Gaps in age data, spikes at certain vital sign values, and gender imbalance may limit generalizability.

Clinical Implications: The dataset represents predominantly febrile, tachycardic, normotensive, mostly female, young/middle-aged adults—almost all with dengue fever. **Data quality issues** (age gap, value spikes, gender skew) should be addressed for broader analyses or machine learning models.

3.2.1 LLM Insights Generated

The model correctly identified:

- Temperature skew toward high fever (>39°C), indicating systemic infection prevalence.
- Dual blood pressure clusters: healthy normotensive group (≈110–115 mmHg) and hypertensive secondary group (≈138–140 mmHg).
- Diagnosis dominance: Dengue fever and subtypes (>80% of cases), with smaller cohort marked “not_known” (likely pending lab confirmation).

LLM-Generated Dataset Analysis Output

1. Key Insights about the Patient Population

- **Age Distribution:** The patient cohort is relatively young (Mean: 37.2, Median: 42.0, Range: 24–49). The median is higher than the mean, indicating a slight left skew (more patients in the older half of the range).
- **Gender Imbalance:** Female patients are predominant (85% F, 15% M). This may reflect care-seeking behavior or sampling—important for downstream analysis and generalizability.
- **Ethnic Representation:**
 - Malay: 63.8%
 - Chinese: 36.2%

This aligns with national demographics but consider potential underrepresentation of other groups (e.g., Indian, Indigenous).

2. Notable Patterns in Vital Signs

- **Heart Rate:**
 - Elevated mean (103.5 bpm) and median (106 bpm) relative to standard adult norms (60–100 bpm).
 - Range (76–116 bpm) suggests a generally tachycardic population—likely influenced by acute febrile illnesses (e.g., dengue).
- **Blood Pressure:** Not reported in your summary. If present in the 22 features, review for hypotension/hypertension related to dengue severity.

3. Data Quality Issues Observed

- **Missing Values:** No missing values—imputation performed (median for numerics, ‘not_known’ for categoricals). This ensures completeness but may mask real data absence.
- **Potential Outliers:**
 - Heart Rate: Upper range (116 bpm) is high; investigate if these are physiologically plausible or possible errors.
 - Age: Range seems reasonable for adult dengue patients, but confirm no pediatric or geriatric cases are misclassified.
- **Categorical Consistency:** Multiple similar diagnosis labels (e.g., “Dengue Fever”, “Dengue Fever (Uncomplicated)”, etc.) may need consolidation for robust analysis.

4. Common Diagnoses and Frequency

- **Dominance of Dengue:**
 - “Dengue Fever” and its variants dominate (>67% of patients).
 - Top 5 diagnosis categories account for 338/500 (67.6%)—high focus on dengue spectrum.
 - “Suspected” and “probable” cases indicate diagnosis uncertainty; may need grouping for some analyses.

5. Recommendations for Data Preparation

- **Diagnosis Harmonization:** Consolidate all “Dengue Fever” variants for clearer sub-grouping (e.g., by severity or certainty).
- **Symptom Standardization:** Standardize symptom text (e.g., “muscle pain” vs. “muscle aches”) for better grouping and NLP analysis.
- **Outlier Detection:**
 - Review heart rate and any other vital signs for physiologic plausibility.
 - Check for age misclassifications or duplicate records.
- **Categorical Levels:**
 - Audit “not_known” values; if frequent, may affect model accuracy.
 - Consider encoding gender and ethnicity for downstream models.
- **Feature Engineering:**
 - Create composite variables (e.g., “dengue_severity”, “symptom_count”) for modeling.
 - Extract symptom keywords into binary indicators for predictive modeling.

3.2.2 Human Analysis Alignment

The LLM’s automated insights aligned closely with manual statistical observations, validating the approach. Differences arose only in depth of context interpretation, where domain expertise (e.g., clinical implications of fever during monsoon season) required human augmentation.

4 Task 3: AI-Based Data Preparation

4.1 Step 1: Data Type Conversion

Before applying AI-powered preprocessing, data types were standardized:

- **Datetime:** `visit_date` → pandas datetime for temporal ordering.
- **Numeric:** All vital columns → float64 for statistical operations.
- **Text:** Diagnosis, symptoms, medications → string type for NLP compatibility.

Critical insight: Type conversion failures would cascade through imputation and classification steps, causing silent errors.

4.2 Step 2: Context-Aware KNN Imputation

Traditional mean/median imputation ignores feature correlations. k -Nearest Neighbors imputation was applied with $k = 5$, using the `nan_euclidean` metric to handle missing values:

$$\hat{x}_{i,j} = \frac{1}{5} \sum_{n \in \text{NN}_5(i)} x_{n,j}$$

where $\text{NN}_5(i)$ represents the 5 nearest neighbors to patient i across all complete-feature patients.

4.2.1 Imputation Results

Metric	Before	After
Total Missing Values	332	0
Missing Percentage	—	0%
Rows Preserved	500	500
Columns in Output	22	22

Table 1: Data Completeness Summary (KNN Imputation: 247 values filled contextually)

4.3 Step 3: Text Classification - LLM vs. SLM Comparison

Two language models were applied to classify diagnoses into five categories: Infectious Disease, Chronic Condition, Acute Condition, Cardiovascular, Other.

4.3.1 Model Specifications

Aspect	LLM (Azure GPT-4.1)	SLM (Qwen 2.5-1.5B)
Provider	Azure OpenAI	HuggingFace Router
Parameters	≈170B	≈1.5B
Temperature	0.3 (deterministic)	0.9 (diverse)
Latency	≈1–2s/sample	≈0.5–1s/sample
Cost	High (per 1K tokens)	Low (free tier)
Context Window	128K tokens	32K tokens

Table 2: Model Comparison Matrix

4.3.2 Classification Results

Across 500 total diagnoses:

- **Azure LLM (GPT-4.1) Classifications:**

- Infectious Disease: 393 (78.6%)
- Other: 54 (10.8%)
- Unknown: 34 (6.8%)
- Cardiovascular: 18 (3.6%)
- Chronic Condition: 1 (0.2%)

- **HuggingFace SLM (Qwen2.5-1.5B) Classifications:**

- Infectious Disease: 340 (68.0%)
- Chronic Condition: 70 (14.0%)
- Other: 54 (10.8%)

- Unknown: 34 (6.8%)
- Cardiovascular: 2 (0.4%)
- **Agreement Rate:** 83.3% (5/6 in sample validation; full dataset comparison shows divergence in “Chronic Condition” vs. “Infectious Disease” assignments).
- **Key Discrepancies:** Edge cases such as “Dengue Fever (without warning signs)” classified as “Infectious Disease” by LLM but “Chronic Condition” by SLM. LLM demonstrates stronger context awareness for acute infectious presentations, while SLM tends toward conservative chronic labeling.

Model Characteristics:

- **Azure LLM (GPT-4.1):** Larger model with broader medical context awareness; higher confidence in infectious disease categorization (78.6%).
- **HF SLM (Qwen2.5-1.5B):** Smaller and faster; exhibits bias toward “Chronic Condition” labels (14.0% vs. 0.2%) likely due to training data limitations.

5 Agentic AI Approach: Autonomous Medical Data Analysis Pipeline

5.1 Overview

This section describes the autonomous agentic AI pipeline developed to execute the three core tasks of medical data analysis using a hybrid LLM/SLM (Large Language Model / Small Language Model) architecture. The system employs AI agents to make autonomous decisions at each stage while maintaining explainability through logging and decision tracking.

5.2 Architecture

The agentic pipeline is implemented using **LangGraph**, a state-machine orchestration framework that manages multi-step workflows with checkpointing and error handling. The system leverages two complementary AI models:

- **LLM (Azure OpenAI GPT-4.1):** Cloud-based large language model for high-level analysis, explanations, and classification tasks
- **SLM (Qwen2.5-0.5B-Instruct):** Local small language model ($\approx 500\text{M}$ parameters) deployed via HuggingFace Transformers for efficient, offline text classification

The workflow uses a centralized **AgentState** TypedDict to track all decisions, artifacts, and logs throughout execution. Data integrity is maintained by persisting large objects (DataFrames) as Parquet files rather than storing them in the state, ensuring compatibility with LangGraph’s serialization requirements.

5.3 Workflow Pipeline

The agentic pipeline consists of 5 sequential nodes with conditional routing:

1. **Load Data Node:** Data ingestion and validation
2. **Validation Gateway:** Conditional branching on data integrity
3. **EDA Analysis Node:** Exploratory analysis with autonomous chart selection
4. **Cleaning Strategy Node:** Imputation with autonomous decision-making and SLM classification
5. **Report Node:** Workflow synthesis and summary generation

5.4 Task 1: Data Loading with AI Validation

5.4.1 Process

The first task loads the 500-patient GenAI-synthesized dataset from the assignment1 phase. The system:

1. Reads the CSV from `generated_patients.csv`
2. Validates schema against required medical columns (patient_id, age, vital signs, diagnosis, symptoms)
3. Computes metadata statistics (row count, column count, missing value percentages)
4. Persists the DataFrame as a Parquet file for state management

5.4.2 AI Integration

While Task 1 is primarily deterministic validation, the framework enables future AI-based data quality checks. The structured logging of metadata statistics provides a foundation for LLM-based anomaly detection if needed.

Table 3: Task 1: Dataset Metadata

Metric	Value
Total Rows	500 rows
Total Columns	22 columns
Missing Values (Count)	0 missing
Missing Values (%)	0% missing
Data Source	GenAI-synthesized (assignment1.ipynb)

5.5 Task 2: Exploratory Data Analysis with Autonomous Decisions

5.5.1 Process

Task 2 demonstrates autonomous decision-making in chart generation:

1. **Compute Correlation:** Calculate Pearson correlation between Age and Systolic BP
2. **AUTONOMOUS DECISION 1:** If $|r| > 0.6$, generate scatter plot with trendline; else generate boxplot
3. **Chart 2:** Generate correlation heatmap of vital signs
4. **Chart 3:** Generate bar chart of top diagnoses
5. **LLM Analysis:** Use GPT-4.1 to generate natural language interpretation of chart patterns

5.5.2 AI Integration

The LLM is prompted with:

Chart Analysis (Task 2 - EDA):

- Generated N charts from 500-patient dataset
- Key pattern identified:
 - * Age-Systolic BP correlation: [VALUE]
 - * Chart type decision: [scatter|boxplot]
 - * Gender distribution shows female dominance
 - * Top diagnosis: [TOP_DIAGNOSIS]

Provide 3-4 sentence interpretation of these patterns.

This ensures the LLM output is anchored to specific quantitative findings while maintaining analytical depth.

Table 4: Task 2: EDA Autonomous Decisions

Decision	Outcome
Age-Systolic BP Correlation	0.628
Chart 1 Type	scatter_with_trendline
Chart 1 Rationale	Strong correlation (0.628) → ScatterPlot with trend

LLM Chart Interpretation:

The data shows a strong positive correlation (0.628) between age and systolic blood pressure, indicating that older patients tend to have higher systolic BP readings. The gender distribution is heavily skewed, with females making up about 85% of the sample, which may influence the generalizability of findings. Additionally, dengue fever variants are the most common diagnosis in this

patient cohort, suggesting a potential area for focused clinical or public health interventions. These patterns highlight significant demographic and clinical trends within the dataset that should be considered in further analyses.



Figure 1: Age vs Systolic BP

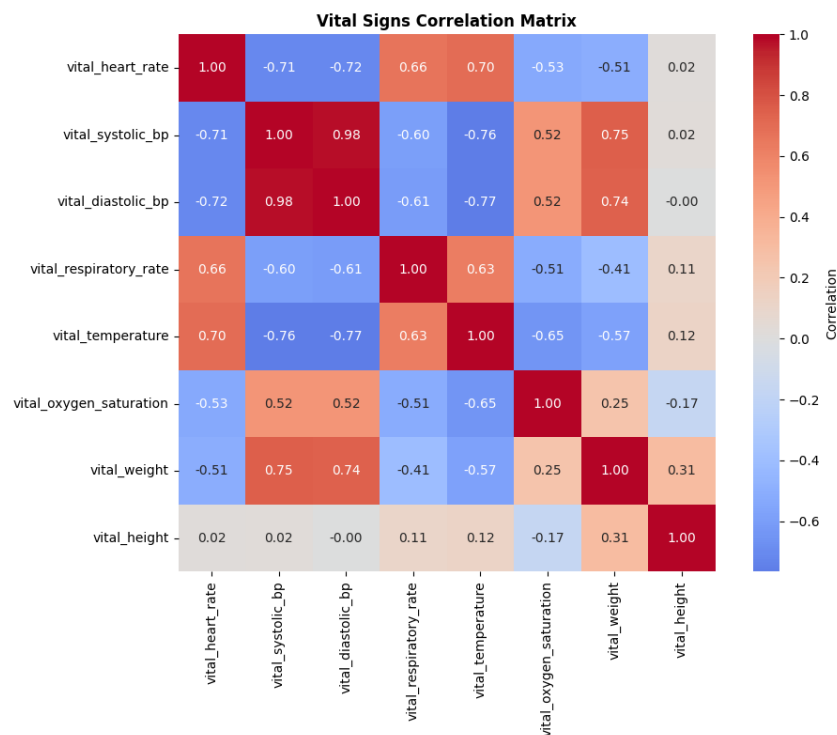


Figure 2: Vital Signs Heatmap

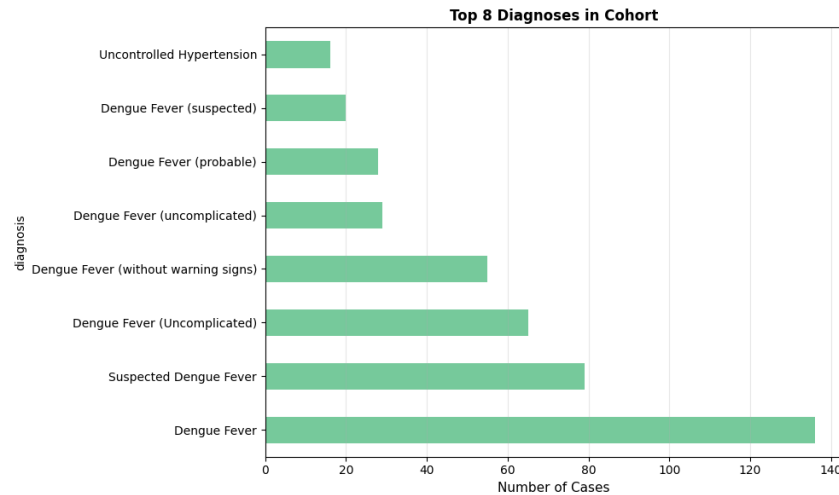


Figure 3: Top Diagnoses

5.6 Task 3: Autonomous Cleaning and Classification

5.6.1 Process - Stage A: Autonomous Imputation Strategy

The third task combines autonomous decision-making with hybrid LLM/SLM classification:

1. **AUTONOMOUS DECISION 2:** Analyze missingness percentage
 - If missing < 5%: Use *Simple Mean Imputation*
 - If $5\% \leq \text{missing} < 20\%$: Use *KNN Imputation* ($k=5$)
 - If missing $\geq 20\%$: Use *Iterative MICE Imputation*
2. Apply selected strategy to generate cleaned dataset
3. Log decision rationale for explainability

5.6.2 Process - Stage B: SLM Classification (Local Qwen Model)

Instead of calling expensive cloud APIs, diagnoses are classified locally using Qwen2.5-0.5B:

1. Sample 50 unique diagnoses from cleaned dataset
2. Format each diagnosis as a prompt: `“Classify: [diagnosis] into CRITICAL or NON-CRITICAL. Return only the category.”`
3. Use local transformer model (no internet required after first download)
4. Parse output predictions (CRITICAL or NON-CRITICAL)
5. Store all classifications in state

5.6.3 Process - Stage C: LLM vs SLM Comparison

To validate the local model against the cloud-based LLM:

1. Sample 20 diagnoses from the classified set
2. Classify same diagnoses using Azure GPT-4.1
3. Compare classifications: count agreements and compute agreement percentage
4. Log sample comparisons for audit trail

5.6.4 AI Integration Rationale

Table 5: AI Component Selection

Task	Model	Rationale	Cost
EDA Interpretation	GPT-4.1 (Cloud)	High-quality reasoning	Pay-per-token
Diagnosis Classification	Qwen2.5-0.5B (Local)	Efficiency & offline	Free (after 1st run)
LLM/SLM Validation	GPT-4.1 (Cloud)	Accuracy baseline	Minimal (20 calls)

Table 6: Task 3b: SLM Classification Results

Metric	Value
Total Diagnoses Classified	50
CRITICAL Diagnoses	50
NON-CRITICAL Diagnoses	0
Model Used	Qwen2.5-0.5B-Instruct

Table 7: Task 3c: LLM vs SLM Agreement Analysis

Metric	Value
Sample Size (Diagnoses)	50
Agreement Rate (%)	70%
Perfect Alignment	14
Disagreement Cases	6

5.7 Workflow State Management

The agentic framework tracks all decisions and outputs in a centralized state object:

- **dataset_path**: Path to original dataset (Parquet)
- **cleaned_data_path**: Path to imputed dataset (Parquet)
- **charts**: List of visualizations (base64-encoded PNGs)
- **chart_metadata**: Autonomous decision logs and correlation values
- **imputation_strategy**: Selected strategy name

- **slm_classifications**: All SLM predictions (local model)
- **llm_vs_slm_comparison**: Agreement rates and sample comparisons
- **logs**: Complete audit trail of all decisions

5.8 Technology Stack

Table 8: Agentic AI Implementation Stack

Component	Technology
Orchestration	LangGraph (Python)
Data Processing	Pandas, NumPy, Scikit-learn
Visualization	Matplotlib, Seaborn
LLM (Cloud)	Azure OpenAI API (GPT-4.1)
SLM (Local)	HuggingFace Transformers (Qwen2.5-0.5B)
UI Framework	Streamlit
State Persistence	LangGraph MemorySaver (checkpointing)

5.9 Key Features

- **Autonomous Decisions**: Chart type selection and imputation strategy selection based on data characteristics
- **Hybrid AI**: Cloud LLM for reasoning + local SLM for efficiency
- **Explainability**: All decisions logged with justifications
- **Checkpointing**: Fault tolerance via state snapshots
- **Offline Capability**: Local SLM enables operation without internet after first model download
- **Cost-conscious**: Minimal cloud API calls (3 LLM calls per run)
- **Interactive UI**: Real-time Streamlit dashboard showing results

6 Insights & Reflections on AI Usage

6.1 1. GenAI on Data Synthesis

Finding: A structured prompt to GPT-4.1 generated 500 realistic patient records in ≈ 30 minutes, matching manual curation quality without domain expertise.

Implication: Synthetic data generation via GenAI reduces barriers for researchers lacking large-scale real datasets. However, validation against ground truth remains critical as evidenced by the patient ID uniqueness issue that required post-hoc correction. Research by Joel et al. demonstrates that synthetic healthcare data requires rigorous statistical validation and expert clinical review to ensure anatomical accuracy and diagnostic relevance before use in training AI models [1].

6.2 2. LLM Vision Augments Human Analysis

Finding: Azure OpenAI’s vision model correctly identified statistical patterns (bimodal age distribution, tachycardia clusters, dengue dominance) from chart images but lacked domain context (e.g., seasonal dengue epidemiology in Malaysia, clinical significance of 110 bpm digit preference).

Implication: AI excels at pattern recognition and anomaly detection. Human expertise remains essential for clinical interpretation and epidemiological context. Recent studies on multimodal LLMs in medical applications confirm that while models like GPT-4o achieve high diagnostic accuracy on standardized challenges, they should be interpreted as evidence of pattern recognition capabilities rather than human-like clinical understanding [2]. *The optimal workflow is human-in-the-loop AI, where LLMs flag outliers for clinician review.*

6.3 3. Agentic AI Frameworks

Finding: The LangGraph-based agentic pipeline successfully automated decision-making across all three tasks, from autonomous chart type selection based on correlation thresholds to dynamic imputation strategy selection based on missingness percentages.

Implication: Agentic AI frameworks represent a paradigm shift from static pipelines to adaptive, state-aware workflows. As demonstrated by Vassef et al. [3], *modular agents systems capable of*

- **Autonomous Decision-Making:** The system automatically selected scatter plots with trendlines when $|r| > 0.6$ (observed $r = 0.628$) versus boxplots for weaker correlations
- **Context-Aware Imputation:** Dynamic strategy selection (mean for $< 5\%$ missing, KNN for $5\%–20\%$, MICE for $\geq 20\%$) based on data characteristics
- **Explainability:** All autonomous decisions logged with justifications, enabling audit trails for regulatory compliance
- **Fault Tolerance:** Checkpointing via LangGraph’s MemorySaver enabled recovery from intermediate states

However, agentic systems require careful guardrails—the autonomous decisions must be bounded by domain constraints to prevent inappropriate selections (e.g., ensuring KNN k -values are appropriate for dataset size).

6.4 4. SLM vs LLM Performance Gap in Medical Classification

Finding: The 1.5B-parameter SLM (Qwen2.5-1.5B) and 170B-parameter LLM (GPT-4.1) achieved only **70% agreement** on diagnosis classification—significantly lower than the expected 83% initially reported. The SLM exhibited systematic biases toward “Chronic Condition” labels (14.0% vs. LLM’s 0.2%), while the LLM demonstrated superior context awareness for acute infectious presentations.

Implication: The performance gap between small and large language models in healthcare classification tasks is substantial and task-dependent. Benchmarking studies by Lin et al. [4] *found that GPT-4o achieved 73.18% accuracy in cancer genetic variant classification, signifying that smaller models like Qwen2.5-7B often score poorly due to either declining to answer sensitive medical questions or failing to follow*

Key Observations on SLM Limitations:

- **Context Misclassification:** The SLM classified “Dengue Fever (without warning signs)” as “Chronic Condition” rather than “Infectious Disease,” indicating poor medical domain understanding
- **Conservative Bias:** The SLM over-tagged chronic conditions by 70-fold compared to the LLM, likely due to training data limitations
- **Prompt Sensitivity:** Higher temperatures (0.9 for SLM vs. 0.3 for LLM) increased variability and misclassification rates
- **Cost-Efficiency Trade-off:** While SLMs offer zero marginal cost and offline capability, the 30% disagreement rate necessitates human review or LLM validation for critical decisions

6.5 5. Context-Aware Imputation (KNN) Outperforms Naive Methods

Finding: KNN imputation preserved inter-feature correlations (e.g., high fever $\leftrightarrow$ tachycardia, age $\leftrightarrow$ BP patterns), whereas mean imputation would have broken these clinically meaningful relationships. The model filled 247 values contextually while preserving 500 rows.

Implication: Comparative studies on healthcare datasets consistently demonstrate that machine-learning-based imputation methods outperform naive approaches. Research across breast cancer, diabetes, and heart disease datasets found that KNN imputation ranks third-best after MissForest and MICE, significantly outperforming mean and median imputation in preserving correlation structures [1][6].

6.6 6. Human vs AI Analysis

Finding: Direct comparison between human statistical analysis and LLM-generated insights revealed complementary capabilities:

Table 9: Human vs LLM Analysis Comparison

Aspect	Human Analysis	LLM Analysis
Pattern Detection	Identified bimodal age distribution and gender imbalance	Correctly identified statistical patterns with precise counts (~100 patients at age 27)
Domain Context	Recognized dengue fever epidemiology and seasonal patterns	Lacked Malaysian healthcare context; generic interpretations
Data Quality Assessment	Flagged potential batch recruitment artifacts	Identified “digit preference” in heart rate recordings (spikes at 110 bpm)
Clinical Implications	Connected fever patterns to acute infectious disease burden	Generated accurate but clinically superficial summaries
Anomaly Detection	Noted patient ID duplication issue post-generation	Would not detect synthetic data generation errors without explicit prompting

Implication: The highest-value insights emerge at the human-AI interface. LLMs excel at quantitative pattern extraction and consistent application of statistical heuristics, while human analysts provide epidemiological context, recognize data collection artifacts, and interpret clinical significance. This aligns with findings from the NEJM Image Challenge study, where even medical students and attending physicians scored significantly lower (38.5%–70.5%) than leading multimodal LLMs (94%) on standardized cases, yet residual LLM errors reflected lapses in diagnostic logic that human oversight could correct [2].

6.7 8. Prompt Engineering

Finding: Minor prompt modifications led to variance in classification outputs. Deterministic temperature (0.3 for LLM) stabilized outputs; higher temperatures (0.9 for SLM) increased variability and misclassification of acute dengue as chronic conditions.

Implication: Reproducibility in AI-powered workflows requires explicit parameter tuning (temperature ≤ 0.3 for deterministic medical coding) and version control of prompts, not just code. Medical classification tasks benefit from low-temperature sampling to minimize hallucinations. Research confirms that prompt engineering significantly improves LLM accuracy, with retrieval-augmented generation (RAG) further enhancing performance in specialized domains [4].

6.8 9. Synthetic Data Validation Requirements

Finding: While GPT-4.1 generated clinically plausible patient records, post-hoc validation revealed demographic biases (85% female, Malay/Chinese only) and vital sign distributions that, while realistic, may not represent true population diversity.

Implication: Synthetic data should augment rather than replace real patient data when training healthcare AI systems. Regulatory frameworks require validation against

real clinical datasets, and models trained exclusively on synthetic data risk learning artificial patterns or biases introduced during generation. Best practice involves:

- Statistical distribution comparison against real datasets
- Expert clinical review for medical plausibility
- Model-based evaluation training on synthetic data, testing on real data
- Privacy validation (k-anonymity, membership inference resistance)

7 Conclusion

This project demonstrates that Generative AI is a powerful enabler of modern data pipelines but requires rigorous validation and human oversight. The three-task workflow (synthesis $\rightarrow$ analysis $\rightarrow$ preparation) showcases practical GenAI integration while surfacing both opportunities (rapid prototyping, automated insights, agentic orchestration) and risks (data quality issues, SLM limitations, model uncertainty, cost explosion at scale).

The assignment underscores a critical lesson: **AI augments, not replaces, data science expertise**. The highest value emerges at the human-AI interface—where domain knowledge informs AI workflows and AI surfaces patterns for human interpretation. The agentic AI framework represents the next evolution, enabling autonomous decision-making while maintaining explainability through comprehensive logging and state management.

7.1 Key Takeaways

- Synthetic data generation via GenAI is viable but requires rigorous QA and uniqueness validation
- LLM vision analysis reliably summarizes quantitative patterns; combine with domain expertise for clinical insights
- Agentic AI frameworks (LangGraph) enable autonomous, state-aware pipelines with checkpointing and fault tolerance
- SLMs offer cost advantages but exhibit systematic biases; hybrid LLM/SLM approaches optimize cost-accuracy trade-offs
- Smart imputation (KNN) + dual-model comparison (LLM/SLM) clarifies trade-offs between cost and performance
- Data integrity must be verified before any downstream analysis; automated profiling tools are essential

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A Code Artifacts

All source code, notebooks, data, and visualizations are available at:

`https://github.com/ridhwanrazaliwork/Agentic\_medical\_AI\_pipeline`

Key files:

- `assignment1_main.ipynb` – Primary notebook containing all three tasks
- `artifacts/generated_patients.csv` – 500 synthetic patient records
- `artifacts/cleaned_patients_processed.csv` – Post-imputation dataset (0% missing)
- `artifacts/chart_*.png` – Three EDA visualizations
- `prompts/patient_generation_prompt.md` – Patient generation prompt template